

Original papers

A robust two-stage framework for maize above-ground biomass prediction integrating spectral remote sensing and allometric growth model

Mohan Yang^{a,1}, Qiang Wu^{a,1}, Jianbo Qi^b, Guijun Yang^{c,d,*}, Zanpu Wang^e, Zhenyu Wang^e, Jun Zhang^a, Hao Yang^{c,*}, Jinpeng Cheng^{a,c,*}, Shuping Xiong^a, Xinming Ma^a^a College of Agronomy, Henan Agricultural University, Zhengzhou 450046, China^b State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China^c Key Laboratory of Quantitative Remote Sensing in Agriculture of Ministry of Agriculture and Rural Affairs, Information Technology Research Center, Beijing Academy of Agriculture and Forestry Sciences, Beijing 100097, China^d College of Geological Engineering and Geomatics, Chang'an University, Xi'an 710054, China^e College of Information and Management Science, Henan Agricultural University, Zhengzhou 450002, China

ARTICLE INFO

Keywords:

Maize
Above-ground biomass (AGB)
Radiative transfer model
Allometric growth model
Spectral remote sensing

ABSTRACT

Above-ground biomass (AGB) is a key indicator for evaluating maize growth dynamics and yield. Although the remote sensing methods have demonstrated utility in biomass estimation, they often overlook the fundamental heterogeneity in spectral contributions between photosynthetic (primarily leaves) and non-photosynthetic organs (stems, ears, and tassels). In this study, we present a methodology to predict AGB by integrating spectral remote sensing and allometric growth theory. We first demonstrate that leaf organs predominate in determining canopy spectral characteristics, with non-leaf components exhibiting minimal influence on spectral signatures. Building on this theoretical foundation, we developed a two-stage estimation framework that first quantifies leaf biomass using canopy spectral indices and subsequently predicts non-leaf organ biomass through stage-specific allometric growth relationships. Results demonstrate the substantial improvements in estimation accuracy, with the framework achieving an R^2 of 0.79 and RMSE of 300.09 g/m². Compared to direct spectral estimation of total AGB, we significantly improve prediction accuracy, demonstrating a 216 % increase in explanatory power and a 46.69 % reduction in error. The framework's robustness across environmental and temporal scales validates its theoretical foundation and practical utility. These findings advance our understanding of biomass allocation dynamics while providing a rigorous approach for non-destructive biomass estimation in maize cultivation systems.

1. Introduction

Above-ground biomass (AGB) encompasses the total weight of plant components above the soil surface, including stems, branches, leaves, flowers, and fruits. AGB is a key ecological indicator for ecosystem analysis, resource evaluation, and climate impact assessment. Its dynamics are governed by water availability, solar radiation, soil nutrients, and species composition (Huang et al., 2016; Vanuytrecht et al., 2014). Traditional destructive AGB sampling methods are inherently resource-intensive and prohibitively costly for large-scale or longitudinal field studies (Jin et al., 2020b; Zhai et al., 2023). Remote sensing provides a non-contact, non-destructive method for rapid large-scale data

acquisition, offering an efficient and objective means for agricultural information collection and analysis (Jin et al., 2020a; Li et al., 2022; Luo et al., 2023).

Advances in quantitative remote sensing have integrated spectral data with machine learning to estimate vegetation AGB. Vegetation indices (VIs) extract vegetation information from spectral data and directly indicate vegetation cover and growth status (Foody et al., 2003; Mourad et al., 2020). The random forest method ensemble algorithm integrates VIs with environmental, biological, and edaphic parameters to analyze high-dimensional datasets, enhancing prediction reliability in ecological modeling (He et al., 2024), and it does not assume a specific data distribution and effectively capture nonlinear relationships

* Corresponding authors at: College of Agronomy, Henan Agricultural University, Zhengzhou 450046, China (J. Cheng).

E-mail addresses: guijun.yang@163.com (G. Yang), younghow@163.com (H. Yang), chengjp@henau.edu.cn (J. Cheng).¹ These authors contributed equally to this work.

between AGB and VIs (Belgiu and Drăguț, 2016; Wu et al., 2023).

Traditional spectral remote sensing methods typically predict overall AGB with good performance (Liu et al., 2018; Xu et al., 2023). However, structural heterogeneity (e.g., leaf aggregation and organ composition), leads to significant differences in spectral responses. In maize crops, the allometric growth of leaf and non-leaf organs varies across phenological stages, with internal structures continuously changing throughout the growth period, making multi-stage AGB monitoring particularly challenging. Zhu et al (2019) estimated total maize AGB by separately assessing above-ground leaf biomass (AGLB) and above-ground stem biomass (AGSB). Yue et al (2023) proposed a multi-stage, multi-organ AGB estimation method for maize, dividing estimation into early vegetative, mid-to-late vegetative, and reproductive stages, and further distinguishing between leaf, stem, and reproductive organs. This separation provides an effective reference for AGB estimation across different growth stages.

Mainstream methods often use the canopy spectral signals to estimate AGB without considering the spectral contributions of different organs (Bastiaanssen and Steduto, 2017; Corti et al., 2018). However, quantifying the spectral contributions of individual plant components (e.g., leaves, stems, ears) through field surveys remains methodologically challenging due to overlapping spectral responses and complex three-dimensional (3D) canopy structures. 3D radiative transfer modeling offers an effective solution. LESS (Large-Scale remote sensing data and image Simulation framework) is widely used in simulating the radiative transfer process in complex scenes, generating large-scale remote sensing images (Qi et al., 2019). By inputting realistic 3D vegetation scenes and spectral properties of various components (stems, leaves, ears, flowers, soil, etc.), high-resolution canopy reflectance can be simulated (Silván-Cárdenas and Corona-Romero, 2017). The construction of 3D scene models and subsequent generation of canopy bidirectional reflectance factor (BRF) data enables quantitative analysis of individual component contributions to canopy spectral characteristics.

The allometric growth model characterizes the differential scaling relationship between an organism's growth rate and its body size or length, providing a quantitative framework extensively utilized in biological, ecological, and interdisciplinary research (Poorter et al., 2012; Zar., 1968). The growth trajectories of organisms frequently exhibit nonlinear scaling relationships with body size, rendering allometric modeling an essential tool for generating predictive frameworks and reference standards in scenes where empirical measurements are unfeasible. Recent studies have utilized allometric models to estimate crop biomass via remote sensing, particularly focusing on maize and wheat (Song et al., 2021; Zhu et al., 2019). However, existing models often rely on plant height as the primary determinant of biomass growth, neglecting the continued biomass increase after height growth ceases, thereby limiting accuracy. Incorporating biomass of dominant organs in canopy spectra as the driving variable for allometric growth may improve spectral remote sensing estimates of maize biomass.

To address these issues, this study proposes a novel framework for predicting maize AGB that integrates a three-dimensional radiative transfer model (3D RTM), a spectral remote sensing estimation model, and an allometric growth model. First, canopy BRF under varying conditions is simulated using the 3D RTM to evaluate contributions of both leaf and non-leaf organs to canopy spectra. Next, a two-step estimation approach is applied: (1) remote sensing estimation of the relationship between canopy VIs and leaf biomass, and (2) allometric growth estimation of the relationship between leaf biomass and non-leaf biomass. This method aims to improve maize AGB estimation accuracy and provide a valuable reference for AGB estimation in other crops.

2. Materials and methods

2.1. Study sites and experimental design

Experiments were conducted at the National Precision Agriculture Research Center in Xiaotangshan, Beijing, China (40.17°N, 116.43°E), with an average elevation of 38 m. The region experiences a semi-humid continental monsoon climate, characterized by an average annual sunshine duration of 2684 h, temperature of 11.8 °C, and precipitation of 500.3 mm. Field experiments spanning 2018–2022 (Fig. 1, Table 1).

2.2. Data collection

2.2.1. Canopy spectral data collection

From 2018 to 2022, canopy spectral data were collected during key maize growth stages (Table 2). Data collection occurred on days with favorable weather conditions (calm or light winds) using UAVs flying between 10:30 AM and 2:30 PM at altitudes of 30–50 m and speeds of 6 m/s, ensuring 80 % overlap and sidelap for comprehensive coverage.

During 2018 to 2019, a Parrot Sequoia multispectral camera mounted on a DJI PHANTOM 4 PRO UAV captured Green (550 nm ± 40 nm), Red (660 nm ± 40 nm), Red-edge (735 nm ± 10 nm), and NIR (790 nm ± 40 nm) bands. The images were processed using Agisoft Metashape Professional software and calibrated following the methodology outlined by Yang et al (Yang et al., 2017a; Yang et al., 2017b).

In 2020, an ASD FieldSpec3 spectroradiometer collected data ranging from 350 to 2500 nm at 1 nm intervals. The sensor was positioned 1 m above the canopy with a 25° field of view. Each sampling point was measured ten times and averaged, with calibration against a standard reference panel conducted before and after each plot's measurements.

During 2021 to 2022, a DJI PHANTOM 4 Multispectral UAV (P4M) equipped with five CMOS sensors captured Blue (450 nm ± 16 nm), Green (560 nm ± 16 nm), Red (650 nm ± 16 nm), Red-edge (730 nm ± 16 nm), and NIR (840 nm ± 26 nm) bands. The images were processed using Terra v2.3.3 (SZ DJI Technology Co., China) for stitching, georeferencing, radiometric calibration, and distortion correction, and were subsequently imported into ENVI 5.6 (Exelis Visual Information Solutions, USA) for extracting band reflectance via the ROI tool.

2.2.2. Data acquisition of above-ground biomass

AGB data were collected concurrently with spectral data through destructive sampling (Table 2). In each plot, 2–4 maize plants were randomly selected, harvested, and separated into stems, leaves, ears, and tassels. Components were dried at 120 °C for 2 h, then at 80 °C until constant weight. Dry weight (DW) was measured using an electronic balance. AGB (g/m²) was calculated using Equation (1). The biomass of leaf organs and non-leaf organs (including stems, ears, and tassels, hereafter referred to as non-leaf biomass) was recorded.

$$AGB = \frac{DW}{PA} \quad (1)$$

where DW is the dry weight per plant (g) and PA is the ground area per plant (m²), calculated as the reciprocal of planting density. Supplementary Table 1 presents descriptive statistics for AGB.

2.3. Evaluating canopy spectral contributions from leaf and non-leaf organs

We developed 3D architectural models of maize plants by using the GroIMP software, including Compact Maize (CM) and Flat-structured Maize (FM). These models were utilized to generate heterogeneous 3D maize population scenes (Fig. 2), each covering a 5 m × 5 m area with randomized plant orientation, a row spacing of 0.6 m, and an intra-row plant spacing of 0.3 m. To evaluate different phenological stages, we simulated 3D plant structures spanning leaf development stages from 9

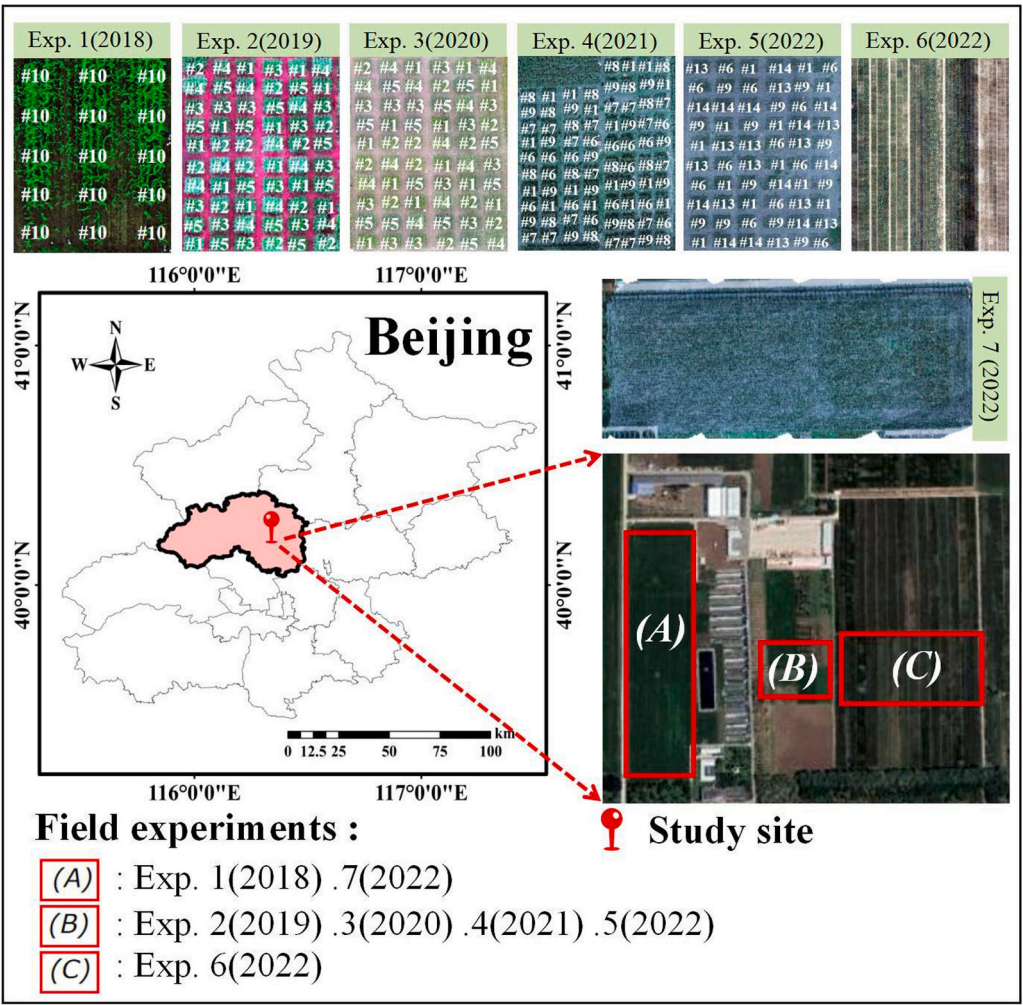


Fig. 1. Geographic locations of study sites and experimental designs. Cultivars: #1 (Zhengdan958), #2 (XianYu335), #3 (Jingnong728), #4 (Chengdan30), #5 (Jingpin6), #6 (Jingjiu16), #7 (Tianci19), #8 (Jingnuo2008), #9 (Nongke336), #10 (Jinghua38), #11 (Yufeng506), #12 (Fengnuo), #13 (Jiyuan1), #14 (Jiyuan168).

Table 1
Experimental treatments on planting density and cultivars over five growing seasons at Xiaotangshan farm.

NO.	Year	Treatments
Exp. 1	2018	Planting density (plants/ha): 8000, 27500, 44000, 64000, 88000; Cultivar: Fig. 1;
Exp. 2	2019	Planting density (plants/ha): 45000, 67500, 90000, 105000; Cultivar: Fig. 1;
Exp. 3	2020	Planting density (plants/ha): 45000, 67500, 90000, 105000; Cultivar: Fig. 1;
Exp. 4	2021	Planting density (plants/ha): 33000, 45000, 67500, 90000; Cultivar: Fig. 1;
Exp. 5	2022	Planting density (plants/ha): 45000, 60000, 67500, 90000; Cultivar: Fig. 1;
Exp. 6	2022	Planting density (plants/ha): 67500; Cultivar: Fig. 1;
Exp. 7	2022	Planting density (plants/ha): 67500; Cultivar: Fig. 1;

to 18 leaves (L9–L18), corresponding to growth phases from jointing to dough stage. Ears and tassels were incorporated into the models starting from the L12 stage.

We used the LESS 3D radiation transfer model (Qi et al., 2019) to simulate the BRF of the maize canopy. The model defines the 3D geometric structure of individual maize organs (stem, leaf, ear), component

Table 2
Spectral data acquisition details.

NO.	Date	Growth stage	Samples	Dataset
Exp. 1(2018)	28th Aug.	Milk	15	Training set
	14th Sep.	Dough	15	
Exp. 2(2019)	12th Sep.	Milk	60	
Exp. 3(2020)	21st Aug.	Blister	60	
Exp. 4(2021)	16th Sep.	Milk	60	Training set
	13th Jul.	Jointing	80	
	28th Jul.	Silking	80	
	10th Aug.	Blister	80	
Exp. 5(2022)	20th Aug.	Milk	80	Test set
	21st Jun.	Jointing	60	
	6th Jul.	Silking	60	
	20th Jul.	Milk	60	
Exp. 6(2022)	6th Aug.	Dough	60	Test set
	21st Jun.	Jointing	5	
Exp. 7(2022)	21st Jul.	Blister	12	Test set
	30th Aug.	Milk	12	
Total of samples			799	

spectral properties, field real scene configuration, and drone observation geometry, and achieves accurate simulation of the canopy BRF in multispectral bands (500–2500 nm), thereby accurately simulating the radiation transfer within the maize canopy. Hemispherical reflectance measurements of maize stems and tassels were conducted under

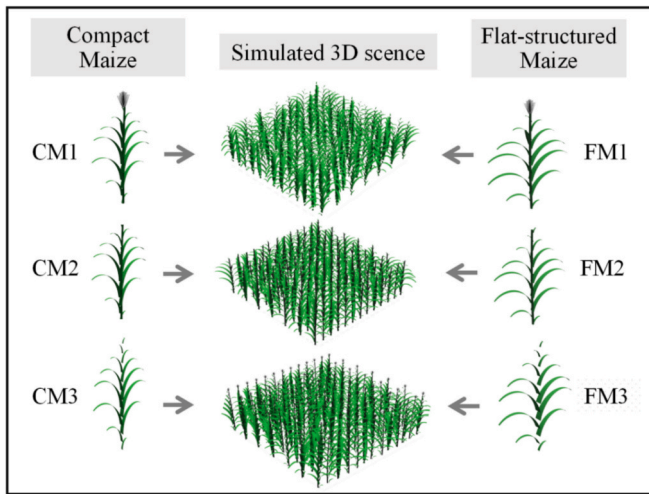


Fig. 2. Example of constructing a 3D model of a maize colony. CM1 represents compact maize with leaves, stem, ear and tassel. CM2 denotes compact maize with leaves and stem. CM3 refers to compact maize with leaves. FM1 represents flat-structured maize with leaves, stem, ear and tassel. FM2 denotes flat-structured maize with leaves, and stem. FM3 refers to flat-structured maize with leaves.

controlled laboratory conditions ($25 \pm 2^\circ\text{C}$, $50 \pm 5\%$ RH) using an ASD FieldSpec®4 Hi-Res NG spectrometer with an integrating sphere accessory. Stem samples (10 cm segments from base, middle, and top sections) were surface-cleaned and positioned flat in fixture grooves fitted with 2 mm rubber O-rings to prevent light leakage. Tassel samples were preprocessed to ensure flat measurement surfaces. Prior to measurements, dark current calibration (with fully closed ports) and white reference calibration (using a Spectralon® standard reflectance panel, 50 % light intensity setting, 150 ms sampling time) were performed. To ensure data quality, 200 scans were averaged per sample to minimize random noise effects, with system recalibration conducted after every five samples to compensate for potential ambient light fluctuations. This measurement protocol yielded complete spectral data spanning 350–2500 nm with a 1 nm output wavelength interval, accurately capturing the hemispherical reflectance characteristics of maize stem and tassel tissues. The hemispherical reflectance of the surface soil (0–5 cm depth) was determined using an integrating sphere system, covering the 350–2500 nm band. The leaf spectra were modeled using the PROSPECT-D algorithm to accurately represent their optical properties, the parameters of PROSPECT-D are clearly presented in Table 3. The input reflectance of each component is shown in Fig. 3.

To evaluate the contributions of leaf and non-leaf organs to canopy spectral reflectance, we compared simulated reflectance between pure leaf scenes (CM3, FM3) and full plant scenes (CM1, FM1). Additionally, we computed ten commonly used VIs (Table 4). The selection of these ten VIs was based on a thorough integration of both theoretical and practical considerations pertinent to the estimation of maize biomass. The Normalized Difference Vegetation Index (NDVI) effectively differentiates vegetation from non-vegetation by exploiting the distinct

Table 3
The parameters of PROSPECT-D.

Parameters	value	Unit
Structure Coefficient	1.8	N
Dry Matter	0.03	Cm-g/cm ²
Brown Pigments	0.0	–
Carotenoids	8.0	Car-μg/cm ²
Chlorophyll	30.0	Cab-μg/cm ²
Anthocyanin	0.0	Anth-μg/cm ²
Water Thickness	0.015	Cw-cm

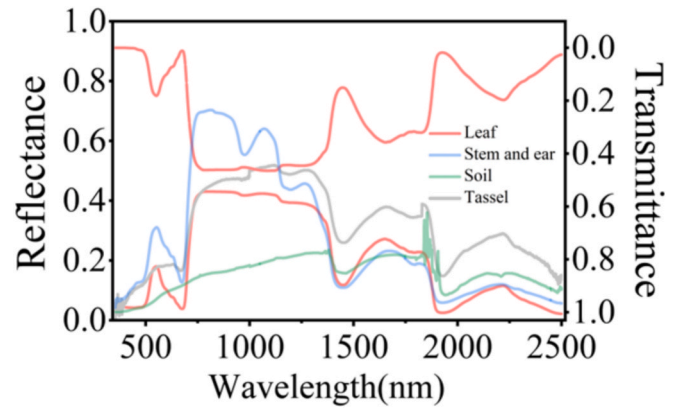


Fig. 3. Component spectral information.

Table 4

Vegetation indices calculation formula.

Vegetation Indices	Formula	Reference
Normalized Difference Vegetation Index	$NDVI = (R_{NIR} - R_{Red}) / (R_{NIR} + R_{Red})$	(Rouse et al., 1974)
Two-Band Enhanced Vegetation Index	$EVI2 = 2.5 \frac{R_{NIR} - R_{Red}}{R_{NIR} + 2.4R_{Red} + 1}$	(Jiang et al., 2008)
Green Chlorophyll Index	$CIGREEN = R_{Red} / R_{Green} - 1$	(Gitelson et al., 2005)
Chlorophyll Index Red Edge	$CIREDEGE = R_{NIR} / R_{RedEdge} - 1$	(Gitelson et al., 2003)
Optimization of Soil-Adjusted Vegetation Index	$OSAVI = \frac{R_{NIR} - R_{Red}}{R_{NIR} + R_{Red} + 0.16}$	(Rondeaux et al., 1996)
Normalized Difference Red Edge	$NDRE = \frac{R_{NIR} - R_{RedEdge}}{R_{NIR} + R_{RedEdge}}$	(Gitelson and Merzlyak, 1994)
Modified Triangular Vegetation Index 2	$MTVI2 = \frac{1.5[1.2(R_{NIR} - R_{Green}) - 2.5(R_{Red} - R_{Green})]}{\sqrt{(2R_{NIR} + 1)^2 - (6R_{NIR} - 5\sqrt{R_{Red}} - 0.5)}}$	(Haboudane et al., 2004)
Green Normalized Difference Vegetation Index	$GNDVI = \frac{R_{NIR} - R_{Green}}{R_{NIR} + R_{Green}}$	(Gitelson et al., 1996)
Modified Chlorophyll Absorption in Reflectance Index	$MCARI = \frac{(R_{RedEdge} - R_{Red}) - 0.2(R_{RedEdge} - R_{Green})}{(R_{RedEdge} / R_{Red})}$	(Daughtry et al., 2000)
Modified Triangular Vegetation Index 1	$MTVI1 = \frac{1.2[1.2(R_{NIR} - R_{Green}) - 2.5(R_{Red} - R_{Green})]}{\sqrt{(2R_{NIR} + 1)^2 - (6R_{NIR} - 5\sqrt{R_{Red}} - 0.5)}}$	(Guan and Liu., 2009)

reflective and absorptive properties of red and near-infrared light. This characteristic makes NDVI widely utilized in agriculture for assessing growth zones and canopy conditions (Wei et al., 2024). Enhanced Vegetation Index 2 (EVI2)(Qiao et al., 2024), Green Chlorophyll Index (GCI)(Nagy et al., 2024), Carotenoid Reflectance Index (CIRE)(Saravia et al., 2022), Optimized Soil-Adjusted Vegetation Index (OSAVI)(Qiao et al., 2022), Normalized Difference Red Edge (NDRE)(Colovic et al., 2022), Modified Triangular Vegetation Index 2 (MTVI2)(Gitelson et al., 2002), Green Normalized Difference Vegetation Index (GNDVI)(Burns et al., 2022), Modified Chlorophyll Absorption in Reflectance Index

(MCARD)(Nagy et al., 2024), and Modified Triangular Vegetation Index 1 (MTVI1)(Jaberi-Aghdam et al., 2024) were selected for their sensitivity to various aspects of vegetation structure and health. These indices improve the model's capacity to capture biomass variability under varying environmental conditions, thereby enhancing estimation accuracy.

2.4. A two-stage framework for maize above-ground biomass prediction

The proposed framework for maize AGB prediction is illustrated in Fig. 4. Initially, the LESS 3D radiative transfer model (RTM) assesses the contributions of leaf and non-leaf organ biomass to the canopy spectrum. The framework estimates maize AGB via a two-stage approach: (1) remote sensing-based estimation of leaf biomass using canopy spectral indices, followed by (2) allometric prediction of non-leaf biomass from leaf biomass.

In the first step, ten commonly used VIs were calculated using canopy spectral reflectance (Table 4). These indices serve as feature variables for estimating leaf organ biomass. A random forest (RF) model is then employed to predict leaf biomass based on these VIs. In the second step, an allometric growth relationship, established across multiple growth stages, correlates AGLB with non-leaf organ biomass in maize (Ledig and Perry., 1966). This relationship is utilized to predict non-leaf organ biomass. The total AGB is subsequently obtained by summing the leaf and non-leaf organ biomass. This study defines traditional remote sensing methods as those directly predicting total AGB using canopy spectral data, whereas our proposed two-stage framework first predicts leaf biomass and then estimates non-leaf biomass through allometric growth relationships.

$$\log M_Y = \log \beta + \alpha \log M_X \quad (2)$$

Where M_Y and M_X represent the masses of organs Y and X, respectively. α is the allometric growth index, β is the proportionality constant.

This study employs experimental field data collected from 2018 to 2021 for model development and optimization (training set). The model

is subsequently validated using 2022 field data, with further validation extended to a large field scale (test set). Both the random forest and allometric growth models are implemented using Python. Model accuracy is evaluated using the coefficient of determination (R^2) and root mean square error (RMSE) as evaluation metrics, defined by the following equations:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2} \quad (4)$$

Where y_i is the actual measured value, $\bar{y}$ is the mean of the measured values, $\hat{y}$ is the predicted value, and n is the number of samples.

3. Results

3.1. Contribution of leaf and non-leaf organs to canopy BRF

Fig. 5 Presents the simulated canopy brf and absolute differences for six maize structures (CM1, CM2, CM3, FM1, FM2, FM3) from 9 to 18 leaves (L9 ~ L18). Canopy spectral reflectance generally increased in both FM and CM structures as the growing season progressed, with differences primarily in the near-infrared (NIR) region. From L9 to L18, FM structures consistently exhibited higher NIR reflectance than CM structures, although this difference diminished at later stages. Absolute differences ($AD|FM1-FM2|$, $AD|FM1-FM3|$, $AD|CM1-CM2|$, $AD|CM1-CM3|$) in canopy reflectance between leaf-only (CM3, FM3) and full-organ (CM1, FM1) simulations remained below 0.012 across all wavelengths (Fig. 5), confirming negligible non-leaf contributions to spectral signatures. This demonstrates that non-leaf organs contribute minimally to canopy spectral reflectance in both compact and flat-structured maize.

Fig. 6 Shows that differences in VIs between leaf-only and full-organ

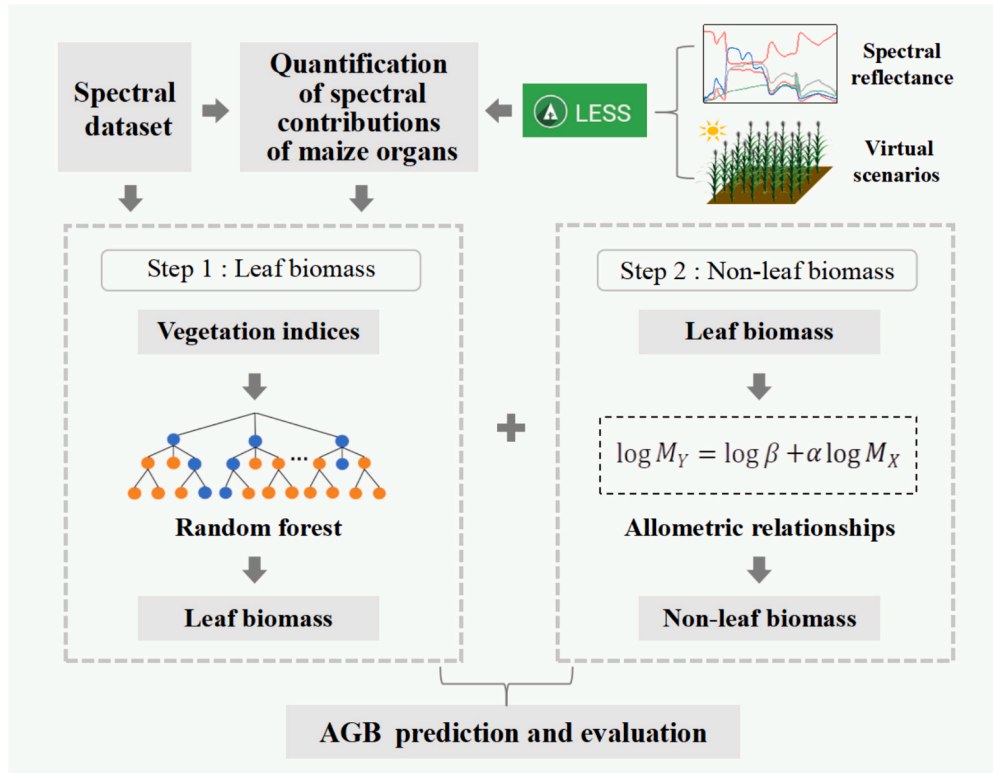


Fig. 4. Overview of the prediction framework.

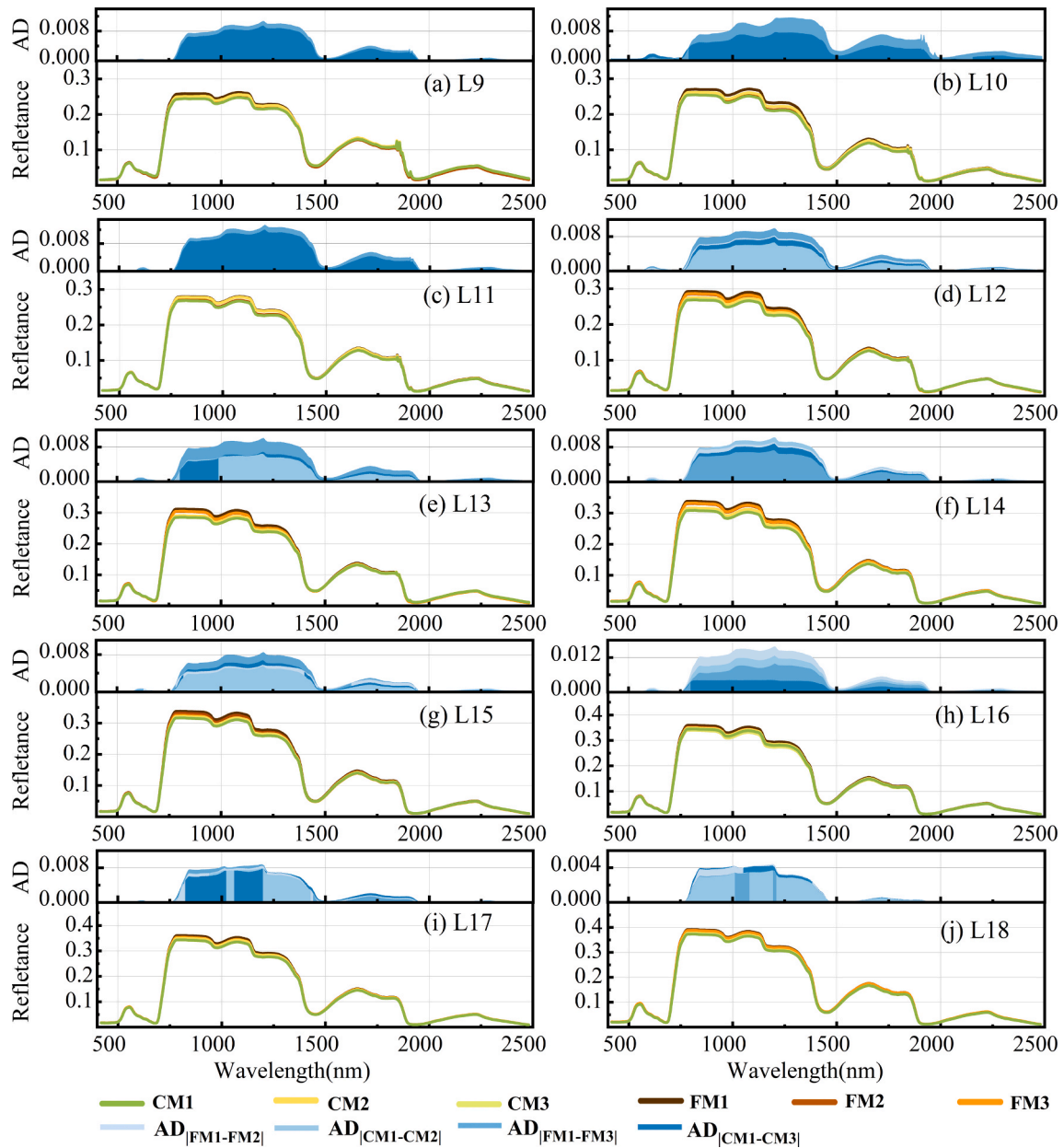


Fig. 5. From (a) to (j), reflectance in 500 nm and 2500 nm bands and its absolute differences are shown in different cases from the 9-leaf stage (L9) to the 18-leaf stage (L18) of maize, where $AD_{| |}$ is the absolute differences between the two simulated canopy reflectance. “CM” represents Compact Maize and “FM” represents Flat-structured Maize. The numbers following these abbreviations denote specific forms of maize: 1 indicates maize that includes leaves, stems, ears, and tassels; 2 means maize containing only leaves and stems; and 3 refers to maize with merely leaves.

populations were minimal, confirming the dominant role of leaf organs in canopy spectra and suggesting that changes in non-leaf organs are not effectively captured by canopy spectral reflectance.

These results underscore the primary role of leaf biomass in modulating canopy spectral signals, providing theoretical justification for leaf-first estimation frameworks.

3.2. Allometric growth relationships between maize leaf and non-Leaf organs biomass

Table 5 presents the allometric models developed for different maize growth stages, with coefficients of determination (R^2) ranging from 0.62 to 0.91. These models demonstrate the dynamic allocation of biomass between leaf and non-leaf organs throughout maize development. The high R^2 values indicate that leaf biomass reliably predicts non-leaf organ

biomass, thereby supporting the use of allometric relationships to enhance the accuracy of maize AGB estimation.

95 % CI represents the lower and upper limits of the 95 % confidence interval.

To further validate the allometric growth model, leaf organ biomass data from the test set were input into the corresponding allometric models for the respective growth stages to predict the biomass of non-leaf organs. These predictions were then compared with the actual measured non-leaf organ biomass in the test set (Fig. 7). The results demonstrated an R^2 of 0.75 and an RMSE of 275.05 g/m², indicating that leaf organ biomass can effectively predict non-leaf organ biomass through the allometric growth model.

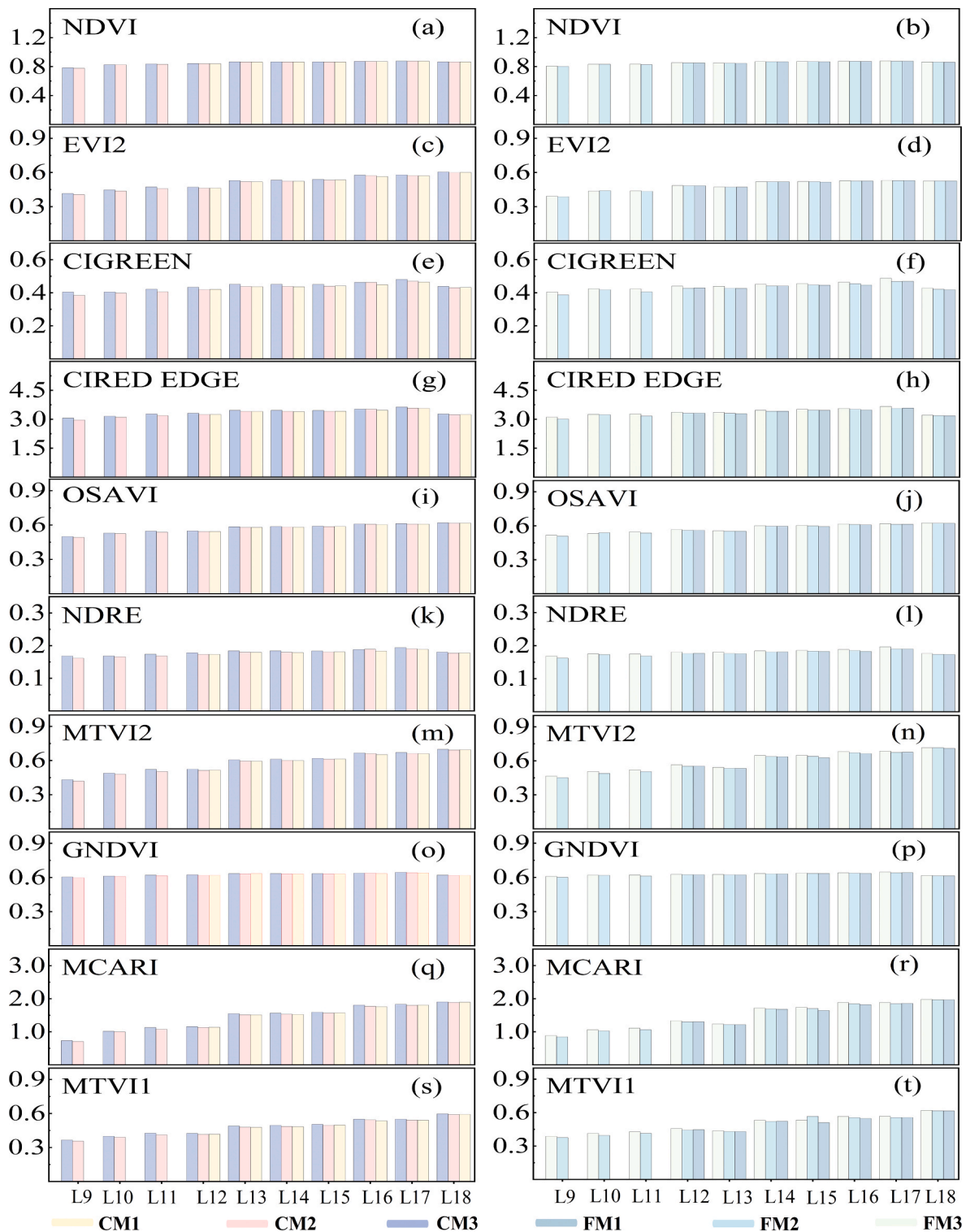


Fig. 6. Vegetation indices were calculated from the simulated reflectance of both leaf-only and full-organ maize populations. The abbreviations “CM” and “FM” followed by numbers (e.g., CM1, CM2) are defined identically to those in Fig. 5.

3.3. Estimation of leaf organ biomass and total above-ground biomass using canopy spectral data

Canopy spectral data demonstrated high accuracy in predicting leaf biomass (Fig. 8b), but showed markedly reduced precision for total AGB estimation (Fig. 8d). The estimation accuracy for maize leaf organ biomass was substantially higher than that for total AGB, achieving an R^2 improvement of 188 %. This indicated a stronger correlation between

leaf organ biomass and canopy spectra, further corroborating the conclusion that canopy spectra are predominantly influenced by leaf organs.

3.4. Performance of the proposed framework for AGB prediction

The proposed two-stage framework demonstrated high R^2 and low RMSE values (Fig. 9), explaining approximately 75 % of the variability

Table 5

Allometric relationship between leaf organ biomass and non-leaf organ biomass during key growth stages of maize.

Growth stage	Allometric model	R ²	95 % CI (g/m ²)
Jointing	$\log M_Y = \log 1.15 + 0.96 \log M_X$	0.90	70.96—81.74
Silking	$\log M_Y = \log 2.98 + 0.89 \log M_X$	0.82	300.79—337.07
Blister	$\log M_Y = \log 0.20 + 1.45 \log M_X$	0.62	481.65—564.44
Milk	$\log M_Y = \log 5.65 + 0.94 \log M_X$	0.73	856.86—941.19
Dough	$\log M_Y = \log 7.00 + 0.85 \log M_X$	0.91	1170.73—1330.01

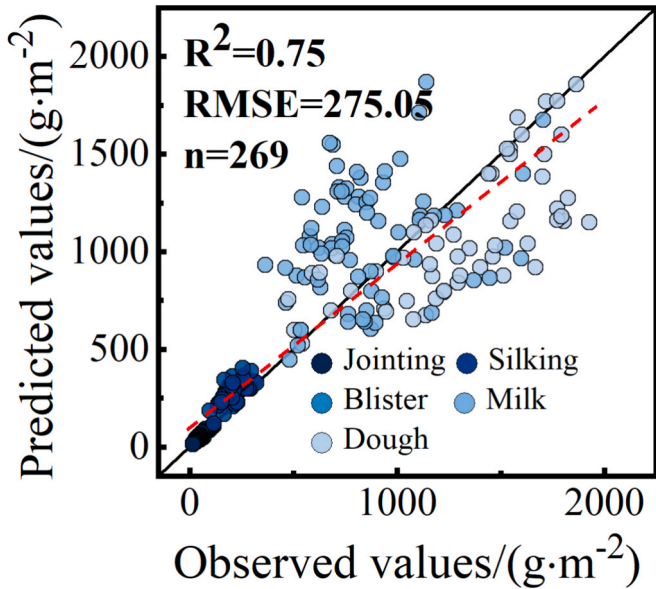


Fig. 7. Performance of the allometric model on the test set.

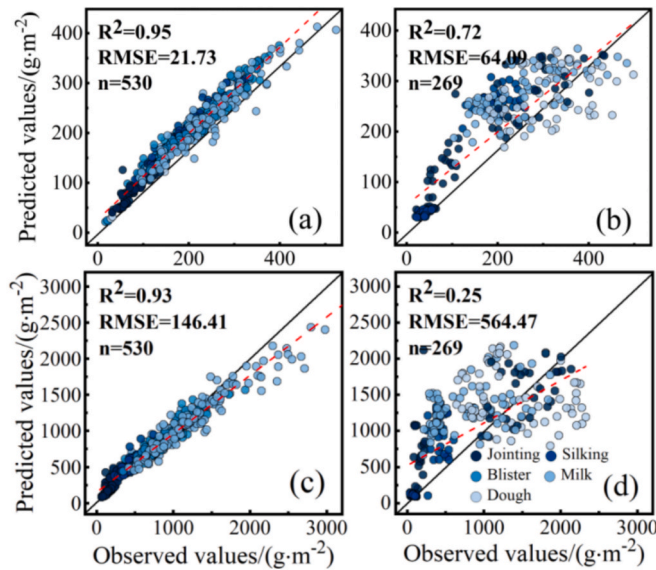


Fig. 8. Canopy spectra were used to estimate leaf organ biomass and AGB of maize. Among them, (a) (b) are the training set and test set performance of VIs estimating leaf organ biomass; (c) (d) are the training set and test set performance of VIs estimating AGB.

in non-leaf biomass and 79 % in total maize AGB. Compared to the direct estimation of total AGB using VIs (Fig. 8d), our framework improved R² by 216 % and reduced RMSE by 46.69 %, confirming the reliability of the two-step estimation method for maize AGB.

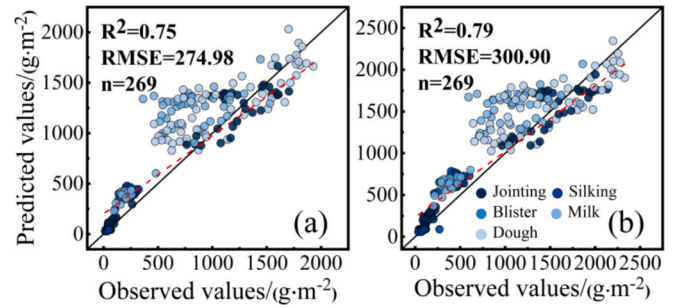


Fig. 9. Estimated performance of the coupled model. Among them, (a) the coupled model estimates the performance of non-leaf organ biomass, and (b) the coupled model estimates the performance of AGB.

Additionally, Fig. 10 demonstrates two key spatiotemporal patterns in AGB accumulation: spatially, planting density drives biomass gradients, with high-density plots (90,000 plants/ha) consistently outperforming low-density plots (45,000 plants/ha), though inter-plant competition partially offsets this advantage post-tasseling; temporally, biomass increases progressively from jointing (June 21) to milk stages (August 6), peaking during grain filling (July 20–August 6) with a 57 % surge (300–1500 g/m²), aligning with maize reproductive dynamics. Our dual-stage framework reliably distinguishes experimental plots from backgrounds and quantifies treatment-specific variations. The consistent spatiotemporal detection validates its robustness for monitoring biomass dynamics during critical growth phases, offering actionable insights for optimizing planting density and field management.

4. Discussion

4.1. Leaf dominance in canopy spectra and implications for model design

The spectral contributions of specific plant organs are inadequately addressed in existing AGB remote sensing models, leading to an over-reliance on leaf-dominant canopy signals (Wang et al., 2022; Yue et al., 2023). The fundamental distinction between traditional direct estimation methods and our two-stage framework lies in separating components directly observable through spectral sensing (leaf characteristics) from those requiring inference through biological relationships (non-leaf biomass). Through LESS 3D simulations, we demonstrated that pure-leaf maize canopies spectrally mimic complete canopies (Figs. 5–6), confirming leaf organ dominance in canopy spectral signatures, with minimal contribution from non-leaf components to reflectance patterns. This finding explains the performance disparity between direct estimation methods for leaf biomass (R² = 0.72, Fig. 8b) and total AGB (R² = 0.25, Fig. 8d).

Our framework transforms traditional empirical models into semi-empirical models by incorporating established biological principles. Traditional approaches, relying solely on statistical correlations between spectral indices and total biomass, frequently fail to generalize across environmental conditions, growing seasons, and developmental stages due to variable spectral contributions from different plant organs. In contrast, our approach applies machine learning exclusively to directly observable components through remote sensing (leaf biomass), while utilizing biologically-grounded allometric relationships for less spectrally active components. The substantial improvements in estimation accuracy (R² increase from 0.25 to 0.79) and error reduction (RMSE decrease from 564.47 to 300.09 g/m²) demonstrate enhanced performance and generalizability of this approach. The framework's robust performance across diverse experimental conditions and growth stages further validates its reliability. By accounting for fundamental heterogeneity in spectral contributions between photosynthetic and non-photosynthetic organs, our methodology provides a mechanistically

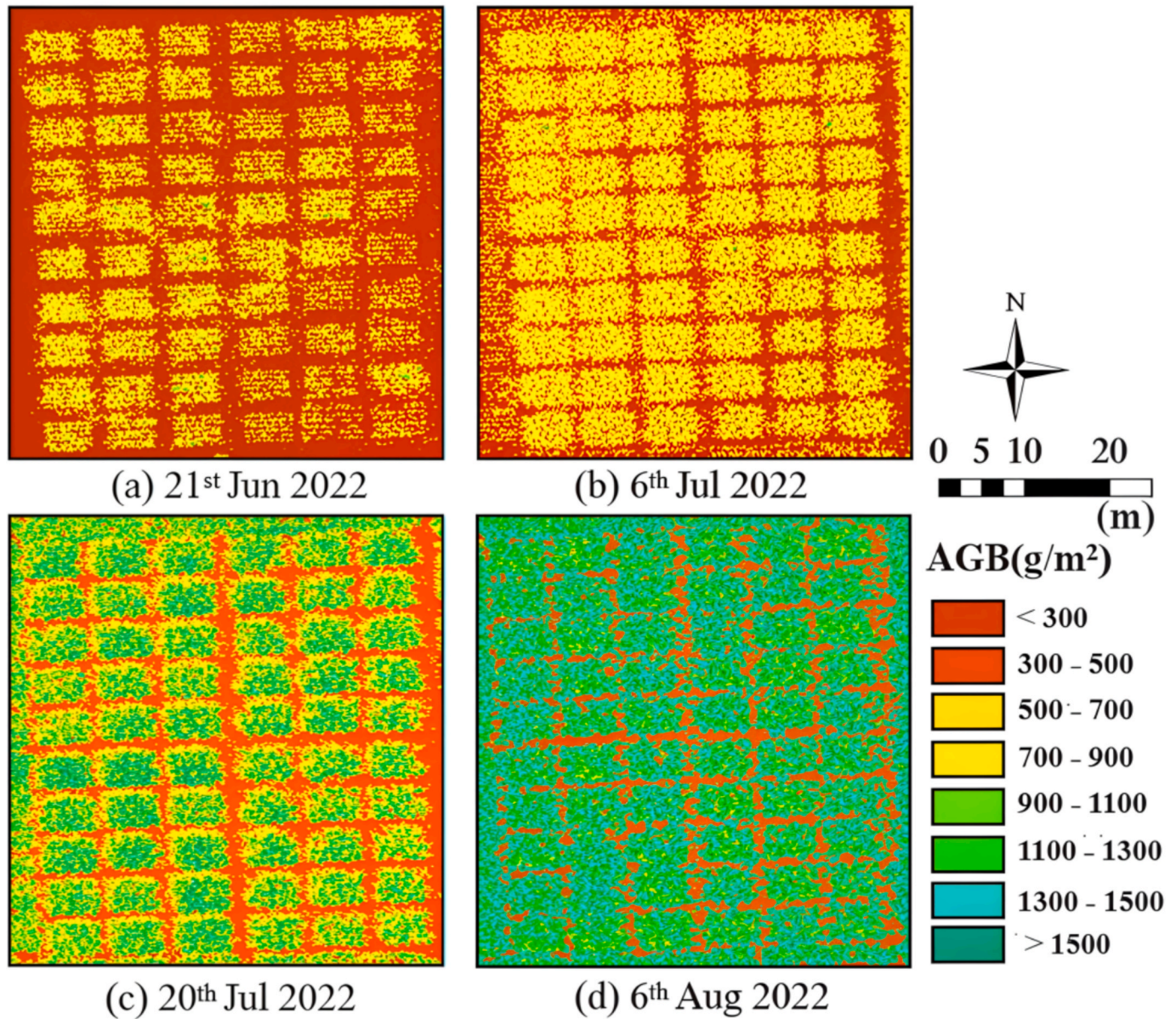


Fig. 10. Spatial distribution map of estimated AGB across experimental plots.

sound approach to AGB estimation that better reflects the biological reality of biomass allocation in maize.

4.2. Biological significance of allometric growth relationships

Allometric growth relationships represent widespread phenomena in biological systems (Huxley et al., 1932; Niklas., 1994). In remote sensing applications, a considerable portion of current research employs morphology-based models (Jucker et al., 2017), while less frequently utilizing potential allometric relationships among plant organs (Ming et al., 2023; Niklas., 2004; Weiner., 2004). This study attempts to apply leaf-to-non-leaf biomass allometry in maize for remote sensing estimation ($R^2 = 0.75$, Fig. 9), suggesting that under specific conditions, this approach may offer potential advantages over certain traditional morphology-driven models. Related research in other disciplines—such as Feitosa et al (2023) on tree biomass allocation and Zhang et al (2021) on maize organ scaling—provides reference for the application of allometric relationships in indirect biomass estimation via remote sensing. The allometric relationships among plant organs may reflect internal coordination mechanisms between different aspects of organ

development (Song et al., 2015). Some studies suggest that under ideal conditions, the allometric growth index might approach 1.0, indicating a relative equilibrium state in plant growth. However, due to changes in growth stages and external environmental factors, this ratio typically fluctuates (Ingestad and Agren., 1991). The relative growth rates and allocation strategies of different maize organs exhibit variations across growth stages. For instance, during the jointing stage, stem growth proceeds relatively rapidly while leaf growth is slower; during the grain-filling stage, leaf growth virtually ceases while grain accumulation accelerates (Zheng et al., 2021).

In this study, our preliminary attempt to establish a unified allometric growth model spanning the entire growth period yielded sub-optimal results. This likely occurred primarily because a unified allometric growth relationship throughout the entire growth period inadequately captures the distinctive growth characteristics of different developmental stages, thereby compromising model estimation accuracy. In contrast, stage-specific modeling appears to more accurately characterize biomass allocation patterns during each critical period. Our results indicate that as the growth period progressed, the coefficient of determination (R^2) for the allometric relationship between leaf and non-

leaf organ biomass displayed a trend of initial decrease followed by subsequent increase, potentially corresponding to variations in plant resource allocation strategies across different growth stages.

4.3. Sample division analysis

Random data splits (e.g., 7:3, 6:4 ratios; Table 6) improved prediction accuracy compared to the baseline model (test R^2 : 0.44–0.45 vs. baseline 0.25). However, precision management in agricultural systems—including irrigation scheduling and fertilization optimization—requires models trained on historical datasets to infer dynamics in unobserved growing seasons, enabling actionable predictions for future cultivation practices.

This framework aligns with challenges documented in agricultural yield forecasting literature, such as “5 years of training + 1 year of validation” (Wang et al., 2025) or historical model calibration (Zhang et al., 2016). or multi-year training with annual independent validation (van Klompenburg et al., 2020). It balances methodological rigor with practical utility, ensuring adaptability to evolving field conditions—including climatic variability and management adjustments—while leveraging longitudinal empirical data to inform evidence-based decision-making.

4.4. Error sources and propagation analysis

The performance differential between our framework’s capacity to estimate leaf biomass ($R^2 = 0.72$, Fig. 8b) versus its ability to directly predict total AGB using traditional methods ($R^2 = 0.25$, Fig. 8d) reveals important insights into error sources and propagation. These can be systematically categorized into three main types:

In the leaf biomass estimation stage, first-stage estimation errors primarily arise from spectral saturation effects, where VIs show reduced sensitivity at higher leaf area index values, particularly during the later growth stages. This phenomenon is evidenced by increased scatter in Fig. 8b at higher biomass values. During the blister and milk stages, leaf senescence begins to impact spectral sensitivity in the red-edge and near-infrared regions, leading to significant deviations in vegetation index-based estimation of leaf biomass. These first-stage errors inevitably propagate into the second stage of the framework.

The allometric growth modeling introduces additional uncertainty, with performance varying significantly across growth stages (Table 5). The lowest coefficient of determination occurs during the blister stage ($R^2 = 0.62$), coinciding with rapid resource reallocation from vegetative to reproductive organs. During this transition, the allometric growth index reaches its highest value ($\alpha = 1.45$), indicating that proportional changes in non-leaf biomass outpace those in leaf biomass. The uncertainties introduced by the allometric model during this stage amplify any estimation errors from the first stage, further complicating the biomass estimation process.

Environmental heterogeneity, including soil properties, nutrient availability, and water retention capacity, significantly modulates biomass accumulation patterns without proportional changes in spectral reflectance (Li et al., 2024a; Rashid et al., 2024; Zhang and Luo., 2024). Variations in management practices such as fertilization regimes and irrigation scheduling introduce additional variability that may not be fully captured by our model (Abdo et al., 2024; Shi et al., 2024). This discrepancy primarily stemmed from insufficient environmental

variability representation in the training dataset, particularly due to the exclusion of critical large-scale field trials (e.g., Exp.6 and Exp.7 in Fig. 1) in the training set, indicating that empirical models for direct AGB estimation from VIs demonstrate poor generalization capacity across different years.

4.5. Understanding non-leaf biomass overestimation

The overestimation of non-leaf biomass and total AGB for certain samples in Fig. 9 by the two-stage framework stems from several interconnected factors. In the blister stage, leaf senescence reduces spectral sensitivity in the red-edge and near-infrared regions, causing deviations in leaf biomass estimation by VIs (test set $R^2 = 0.72$)—errors that propagate through the allometric growth model. Relying solely on leaf biomass as a driver cannot capture the sudden growth of reproductive organs, ultimately leading to AGB overestimation. During reproductive stages, substantial biomass accumulation occurs in non-leaf organs that contribute minimally to canopy spectral signatures, as confirmed by our LESS simulation results. While canopy VIs like NDVI typically increase and then decrease throughout the growing season (Shuai et al., 2023), total AGB continues to rise (Qi et al., 2020), creating an inherent disconnect between spectral signatures and biomass accumulation patterns. This phenomenon is particularly pronounced during the blister and milk stages when ear development accelerates without proportional changes in leaf biomass or spectral reflectance.

Despite these error sources, our framework achieves significantly improved overall performance ($R^2 = 0.79$, Fig. 9b) compared to direct estimation methods. This suggests that the biological constraints imposed by the allometric relationships effectively mitigate error propagation, particularly during growth stages with more stable biomass allocation patterns (jointing, $R^2 = 0.90$; dough, $R^2 = 0.91$).

4.6. Practical applications and future directions

The framework for precision agricultural practices proposed in this study effectively guides real-world agricultural management decisions (Li et al., 2024b). Specifically, it assists with: nitrogen application timing and rates during vegetative growth; irrigation frequency determination based on biomass partitioning patterns during the critical silking-to-grain-filling transition; and optimization of foliar nutrient applications through differentiation between normal senescence and stress-induced alterations during later reproductive phases. Furthermore, the framework’s capacity to monitor organ-specific biomass allocation facilitates density management decisions, enabling farmers to modulate plant populations according to biomass distribution patterns and resource availability.

Future refinements should focus on optimizing transitional-stage models through growth-stage-specific calibration of allometric parameters, organ-specific VIs, and high-frequency temporal sampling to capture dynamic biomass shifts. This requires integrating multi-source remote sensing (e.g., LiDAR for structural phenotyping) and environmental covariates (accumulated temperature, soil data) to build spatiotemporal dynamic models, enhancing prediction accuracy during critical phenological transitions.

5. Conclusions

This study utilized the LESS 3D radiative transfer model to demonstrate that maize canopy spectra are primarily influenced by leaf organs. We developed a novel two-stage framework for predicting maize AGB, achieving high accuracy with an R^2 of 0.79 and RMSE of 300.09 g/m². The framework first estimates leaf biomass using VIs and then predicts non-leaf biomass through allometric growth relationships, thereby enhancing AGB estimation accuracy and reliability. Notably, it shows significant generalization improvements across different years and field conditions, providing a robust tool for agricultural monitoring and

Table 6
Sample division results.

Training Set	Test Set	Total Above-Ground Biomass (R^2)	
		Training	Test
70 % of all the data	The remaining 30 %	0.93	0.45
60 % of all the data	The remaining 40 %	0.93	0.44

management.

This framework circumvents time-consuming and labor-intensive AGB measurement methods while achieving higher estimation accuracy. By integrating advanced radiative transfer modeling and allometric growth relationships, this method offers broad applicability for AGB estimation in other crops. It provides a comprehensive and efficient solution for precise biomass estimation and enhanced agricultural resource management.

CRedit authorship contribution statement

Mohan Yang: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Conceptualization. **Qiang Wu:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis. **Jianbo Qi:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis. **Guijun Yang:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Zanpu Wang:** Software, Formal analysis. **Zhenyu Wang:** Validation, Data curation. **Jun Zhang:** Validation, Software, Methodology. **Hao Yang:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Jinpeng Cheng:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Shuping Xiong:** Writing – review & editing, Supervision. **Xinming Ma:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the Natural Science Foundation of China (32401705, 42371373), the National Key Research and Development Program of China (2024YFD2301100), the China Postdoctoral Science Foundation (GZC20230721), Key Technology Research and Development Program of Henan Province (242102111192), the Key Research and Development Program of Henan Province (251111111000).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2025.110398>.

Data availability

The authors do not have permission to share data.

References

- Abdo, A.I., Sun, D., Shi, Z., Abdel-Fattah, M.K., Zhang, J., Kuzyakov, Y., 2024. Conventional agriculture increases global warming while decreasing system sustainability. *Nat. Clim. Chang.* 1–8.
- Bastiaanssen, W.G.M., Steduto, P., 2017. The water productivity score (WPS) at global and regional level: Methodology and first results from remote sensing measurements of wheat, rice and maize. *Sci. Total Environ.* 575, 595–611.
- Belgiu, M., Drăguț, L., 2016. Random forest in remote sensing: A review of applications and future directions. *ISPRS J. Photogramm. Remote Sens.* 114, 24–31.
- Burns, B.W., Green, V.S., Hashem, A.A., Massey, J.H., Shew, A.M., Adviento-Borbe, M.A., Milad, M., 2022. Determining nitrogen deficiencies for maize using various remote sensing indices. *Precis. Agric.* 23, 791–811.
- Colovic, M., Yu, K., Todorovic, M., Cantore, V., Hamze, M., Albrizio, R., Stellacci, A.M., 2022. Hyperspectral vegetation indices to assess water and nitrogen status of sweet maize crop. *Agronomy* 12, 2181.
- Corti, M., Cavalli, D., Cabassi, G., Marino Gallina, P., Bechini, L., 2018. Does remote and proximal optical sensing successfully estimate maize variables? A review. *Eur. J. Agron.* 99, 37–50.
- Daughtry, C.S., Walthall, C., Kim, M., De Colstoun, E.B., McMurtrey III, J., 2000. Estimating corn leaf chlorophyll concentration from leaf and canopy reflectance. *Remote Sens. Environ.* 74, 229–239.
- Feitosa, T.S., de Carvalho, E.C.D., Barreto, R.W., Mantovani, W., de Araújo, F.S., da Costa, R.C., 2023. Use of support influences height and above-ground allometry but not biomass allocation to different aerial organs of an invasive vine. *Trees* 37, 373–383.
- Food, G.M., Boyd, D.S., Cutler, M.E., 2003. Predictive relations of tropical forest biomass from Landsat TM data and their transferability between regions. *Remote Sens. Environ.* 85, 463–474.
- Gitelson, A., Merzlyak, M.N., 1994. Quantitative estimation of chlorophyll-a using reflectance spectra: Experiments with autumn chestnut and maple leaves. *J. Photochem. Photobiol. B Biol.* 22, 247–252.
- Gitelson, A.A., Gritz, Y., Merzlyak, M.N., 2003. Relationships between leaf chlorophyll content and spectral reflectance and algorithms for non-destructive chlorophyll assessment in higher plant leaves. *J. Plant Physiol.* 160, 271–282.
- Gitelson, A.A., Kaufman, Y.J., Merzlyak, M.N., 1996. Use of a green channel in remote sensing of global vegetation from EOS-MODIS. *Remote Sens. Environ.* 58, 289–298.
- Gitelson, A.A., Kaufman, Y.J., Stark, R., Rundquist, D., 2002. Novel algorithms for remote estimation of vegetation fraction. *Remote Sens. Environ.* 80, 76–87.
- Gitelson, A.A., Viña, A., Ciganda, V., Rundquist, D.C., Arkebauer, T.J., 2005. Remote estimation of canopy chlorophyll content in crops. *Geophys. Res. Lett.* 32.
- Guan, L., Liu, X., 2009. Two Kinds of Modified Spectral Indices for Retrieval of Crop Canopy Chlorophyll Content. *Adv. Earth Science* 24, 548.
- Haboudane, D., Miller, J.R., Pattey, E., Zarco-Tejada, P.J., Strachan, I.B., 2004. Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture. *Remote Sens. Environ.* 90, 337–352.
- He, Z., Wang, J., Jiang, M., Hu, L., Zou, Q., 2024. Random subsequence forests. *Inf. Sci.* 667, 120478.
- Huang, J., Sedano, F., Huang, Y., Ma, H., Li, X., Liang, S., Tian, L., Zhang, X., Fan, J., Wu, W., 2016. Assimilating a synthetic Kalman filter leaf area index series into the WOFOST model to improve regional winter wheat yield estimation. *Agric. For. Meteorol.* 216, 188–202.
- Huxley, J.S., Churchill, F.B., Strauss, R.E., 1932. Problems of relative growth. Johns Hopkins University Press.
- Ingestad, T., Agren, G.L., 1991. The influence of plant nutrition on biomass allocation. *Ecol. Appl.* 1, 168–174.
- Jaberi-Aghdam, M., Momayezi, M.R., Bagheri, N., Azizi, P., Nasri, M., 2024. Nitrogen assessment by remote sensing and multispectral imagery in maize (*Zea mays* L.). *J. Crop Sci. Biotechnol.* 27, 31–41.
- Jiang, Z., Huete, A.R., Didan, K., Miura, T., 2008. Development of a two-band enhanced vegetation index without a blue band. *Remote Sens. Environ.* 112, 3833–3845.
- Jin, S., Su, Y., Song, S., Xu, K., Hu, T., Yang, Q., Wu, F., Xu, G., Ma, Q., Guan, H., Pang, S., Li, Y., Guo, Q., 2020a. Non-destructive estimation of field maize biomass using terrestrial lidar: an evaluation from plot level to individual leaf level. *Plant Methods* 16, 69.
- Jin, X., Li, Z., Feng, H., Ren, Z., Li, S., 2020b. Deep neural network algorithm for estimating maize biomass based on simulated Sentinel 2A vegetation indices and leaf area index. *The Crop Journal* 8, 87–97.
- Jucker, T., Caspersen, J., Chave, J., Antin, C., Barbier, N., Bongers, F., Dalponte, M., van Ewijk, K.Y., Forrester, D.L., Haeni, M., 2017. Allometric equations for integrating remote sensing imagery into forest monitoring programmes. *Glob. Chang. Biol.* 23, 177–190.
- Ledig, F.T., & Perry, T.O. (1966). Physiological genetics of the shoot-root ratio.
- Li, Q., Jin, S., Zang, J., Wang, X., Sun, Z., Li, Z., Xu, S., Ma, Q., Su, Y., Guo, Q., Jiang, D., 2022. Deciphering the contributions of spectral and structural data to wheat yield estimation from proximal sensing. *The Crop Journal* 10, 1334–1345.
- Li, S., Sun, Y., Zhang, L., Jiang, X., Yu, H., 2024a. Heterogeneous interface engineering of bionic corn-structured ternary nanocomposites for excellent low-frequency microwave absorption. *Mater. Today Phys.* 42, 101390.
- Li, Y., Li, C., Cheng, Q., Duan, F., Zhai, W., Li, Z., Mao, B., Ding, F., Kuang, X., Chen, Z., 2024b. Estimating Maize Crop Height and Aboveground Biomass Using Multi-Source Unmanned Aerial Vehicle Remote Sensing and Optuna-Optimized Ensemble Learning Algorithms. *Remote Sens. (Basel)* 16, 3176.
- Liu, S., Yang, Y., Shen, H., Hu, H., Zhao, X., Li, H., Liu, T., Fang, J., 2018. No significant changes in topsoil carbon in the grasslands of northern China between the 1980s and 2000s. *Sci. Total Environ.* 624, 1478–1487.
- Luo, L., Sun, S., Xue, J., Gao, Z., Zhao, J., Yin, Y., Gao, F., Luan, X., 2023. Crop yield estimation based on assimilation of crop models and remote sensing data: A systematic evaluation. *Agr. Syst.* 210, 103711.
- Ming, H., Cai, F., Mi, N., Zhang, Y., Zhang, H., Zhang, S., Zhao, X., Feng, Y., 2023. Allometric Relationships between Dry Matter Weights of Maize Organs and Their Responses to Drought. *Water* 15, 99.
- Mourad, R., Jaafar, H., Anderson, M., Gao, F., 2020. Assessment of leaf area index models using harmonized landsat and sentinel-2 surface reflectance data over a semi-arid irrigated landscape. *Remote Sens. (Basel)* 12, 3121.
- Nagy, A., Szabó, A., Elbeltagi, A., Nxumalo, G.S., Bódi, E.B., Tamás, J., 2024. Hyperspectral indices data fusion-based machine learning enhanced by MRMR algorithm for estimating maize chlorophyll content. *Front. Plant Sci.* 15, 1419316.
- Niklas, K.J., 1994. Plant allometry: the scaling of form and process. University of Chicago Press.

- Niklas, K.J., 2004. Plant allometry: is there a grand unifying theory? *Biol. Rev.* 79, 871–889.
- Poorter, H., Niklas, K.J., Reich, P.B., Oleksyn, J., Poot, P., Mommer, L., 2012. Biomass allocation to leaves, stems and roots: meta-analyses of interspecific variation and environmental control. *New Phytol.* 193, 30–50.
- Qi, D., Hu, T., Liu, T., 2020. Biomass accumulation and distribution, yield formation and water use efficiency responses of maize (*Zea mays* L.) to nitrogen supply methods under partial root-zone irrigation. *Agric Water Manag* 230, 105981.
- Qi, J., Xie, D., Yin, T., Yan, G., Gastellu-Etchegorry, J.-P., Li, L., Zhang, W., Mu, X., Norford, L.K., 2019. LESS: Large-Scale remote sensing data and image simulation framework over heterogeneous 3D scenes. *Remote Sens. Environ.* 221, 695–706.
- Qiao, K., Zhu, W., Xie, Z., Wu, S., Li, S., 2024. New three red-edge vegetation index (VI3RE) for crop seasonal LAI prediction using Sentinel-2 data. *Int. J. Appl. Earth Obs. Geoinf.* 130, 103894.
- Qiao, L., Zhao, R., Tang, W., An, L., Sun, H., Li, M., Wang, N., Liu, Y., Liu, G., 2022. Estimating maize LAI by exploring deep features of vegetation index map from UAV multispectral images. *Field Crop Res* 289, 108739.
- Rashid, R., Aslam, W., Aziz, R., Aldehim, G., 2024. An early and smart detection of corn plant leaf diseases using IoT and deep learning multi-models. *IEEE Access* 12, 23149–23162.
- Rondeaux, G., Steven, M., Baret, F., 1996. Optimization of soil-adjusted vegetation indices. *Remote Sens. Environ.* 55, 95–107.
- Rouse, J.W., Jr., Haas, R.H., Deering, D., Schell, J., Harlan, J.C., 1974. Monitoring the vernal advancement and retrogradation (green wave effect) of natural vegetation. E75-10354. 1974.
- Saravia, D., Salazar, W., Valqui-Valqui, L., Quille-Mamani, J., Porras-Jorge, R., Corredor, F.-A., Barboza, E., Vásquez, H.V., Casas Díaz, A.V., Arbizu, C.I., 2022. Yield predictions of four hybrids of maize (*Zea mays*) using multispectral images obtained from UAV in the coast of Peru. *Agronomy* 12, 2630.
- Shi, T.-S., Collins, S.L., Yu, K., Peñuelas, J., Sardans, J., Li, H., Ye, J.-S., 2024. A global meta-analysis on the effects of organic and inorganic fertilization on grasslands and croplands. *Nat. Commun.* 15, 3411.
- Shuai, G., Fowler, A., Basso, B., 2023. Within-season vegetation indices and yield stability as a predictor of spatial patterns of Maize (*Zea mays* L.) yields. *Precis. Agric.* 1–20.
- Silván-Cárdenas, J., Corona-Romero, N., 2017. Radiation budget of vegetation canopies with reflective surface: A generalization using the Markovian approach. *Remote Sens. Environ.* 189, 118–131.
- Song, Y., Birch, C., Rui, Y., Hanan, J., 2015. Allometric relationships of maize organ development under different water regimes and plant densities. *Plant Prod. Sci.* 18, 1–10.
- Song, Y., Wang, J., Shan, B., 2021. Estimation of winter wheat yield from UAV-based multi-temporal imagery using crop allometric relationship and SAFY model. *Drones* 5, 78.
- van Klompenburg, T., Kassahun, A., Catal, C., 2020. Crop yield prediction using machine learning: A systematic literature review. *Comput. Electron. Agric.* 177, 105709.
- Vanuytrecht, E., Raes, D., Steduto, P., Hsiao, T.C., Fereres, E., Heng, L.K., Vila, M.G., Moreno, P.M., 2014. AquaCrop: FAO's crop water productivity and yield response model. *Environ. Model. Softw.* 62, 351–360.
- Wang, F., Yang, M., Ma, L., Zhang, T., Qin, W., Li, W., Zhang, Y., Sun, Z., Wang, Z., Li, F., 2022. Estimation of above-ground biomass of winter wheat based on consumer-grade multi-spectral UAV. *Remote Sens. (Basel)* 14, 1251.
- Wang, Y., Yu, Q., Liu, Z., Ren, W., Lu, X., 2025. A practical SIF-based crop model for predicting crop yields by quantifying the fraction of open PSII reaction centers (qL). *Remote Sens. Environ.* 320, 114658.
- Wei, Y., Lu, M., Yu, Q., Li, W., Wang, C., Tang, H., Wu, W., 2024. The normalized difference yellow vegetation index (NDYVI): A new index for crop identification by using GaoFen-6 WFV data. *Comput. Electron. Agric.* 226, 109417.
- Weiner, J., 2004. Allocation, plasticity and allometry in plants. *Perspect. Plant Ecol. Evol. Syst.* 6, 207–215.
- Wu, Q., Zhang, Y., Zhao, Z., Xie, M., Hou, D., 2023. Estimation of relative chlorophyll content in spring wheat based on multi-temporal UAV remote sensing. *Agronomy* 13, 211.
- Xu, T., Wang, F., Shi, Z., Xie, L., Yao, X., 2023. Dynamic estimation of rice aboveground biomass based on spectral and spatial information extracted from hyperspectral remote sensing images at different combinations of growth stages. *ISPRS J. Photogramm. Remote Sens.* 202, 169–183.
- Yang, G., Li, C., Wang, Y., Yuan, H., Feng, H., Xu, B., Yang, X., 2017a. The DOM generation and precise radiometric calibration of a UAV-mounted miniature snapshot hyperspectral imager. *Remote Sens. (Basel)* 9, 642.
- Yang, G., Liu, J., Zhao, C., Li, Z., Huang, Y., Yu, H., Xu, B., Yang, X., Zhu, D., Zhang, X., 2017b. Unmanned aerial vehicle remote sensing for field-based crop phenotyping: current status and perspectives. *Front. Plant Sci.* 8, 1111.
- Yue, J., Yang, H., Yang, G., Fu, Y., Wang, H., Zhou, C., 2023. Estimating vertically growing crop above-ground biomass based on UAV remote sensing. *Comput. Electron. Agric.* 205, 107627.
- Zar, J.H., 1968. Calculation and miscalculation of the allometric equation as a model in biological data. *Bioscience* 18, 1118–1120.
- Zhai, W., Li, C., Fei, S., Liu, Y., Ding, F., Cheng, Q., Chen, Z., 2023. CatBoost algorithm for estimating maize above-ground biomass using unmanned aerial vehicle-based multi-source sensor data and SPAD values. *Comput. Electron. Agric.* 214, 108306.
- Zhang, L., Guo, C.L., Zhao, L.Y., Zhu, Y., Cao, W.X., Tian, Y.C., Cheng, T., Wang, X., 2016. Estimating wheat yield by integrating the WheatGrow and PROSAIL models. *Field Crop Res* 192, 55–66.
- Zhang, X.-F., Luo, C.-L., Mo, F., Ren, H.-X., Mburu, D., Kavagi, L., Dai, R.-Z., Wesly, K., Ren, A.-T., Nyende, A.B., Xiong, Y.-C., 2021. Density-dependent maize (*Zea mays* L.) yield increase in trade-off in reproductive allocation and water use under ridge-furrow plastic-mulching. *Field Crop Res* 264, 108102.
- Zhang, Z., Luo, W., 2024. Preparation of amidogen-rich monolithic corn bract via surface graft modification for efficient separation of acid red from wastewater. *Chem. Pap.* 78, 4827–4839.
- Zheng, B., Guo, X., Tang, Y., Chen, L., Xie, F., 2021. Development changes in multi-scale structure and functional properties of waxy corn starch at different stages of kernel growth. *Int. J. Biol. Macromol.* 191, 335–343.
- Zhu, Y., Zhao, C., Yang, H., Yang, G., Han, L., Li, Z., Feng, H., Xu, B., Wu, J., Lei, L., 2019. Estimation of maize above-ground biomass based on stem-leaf separation strategy integrated with LiDAR and optical remote sensing data. *PeerJ* 7, e7593.